

Supplementary Figures

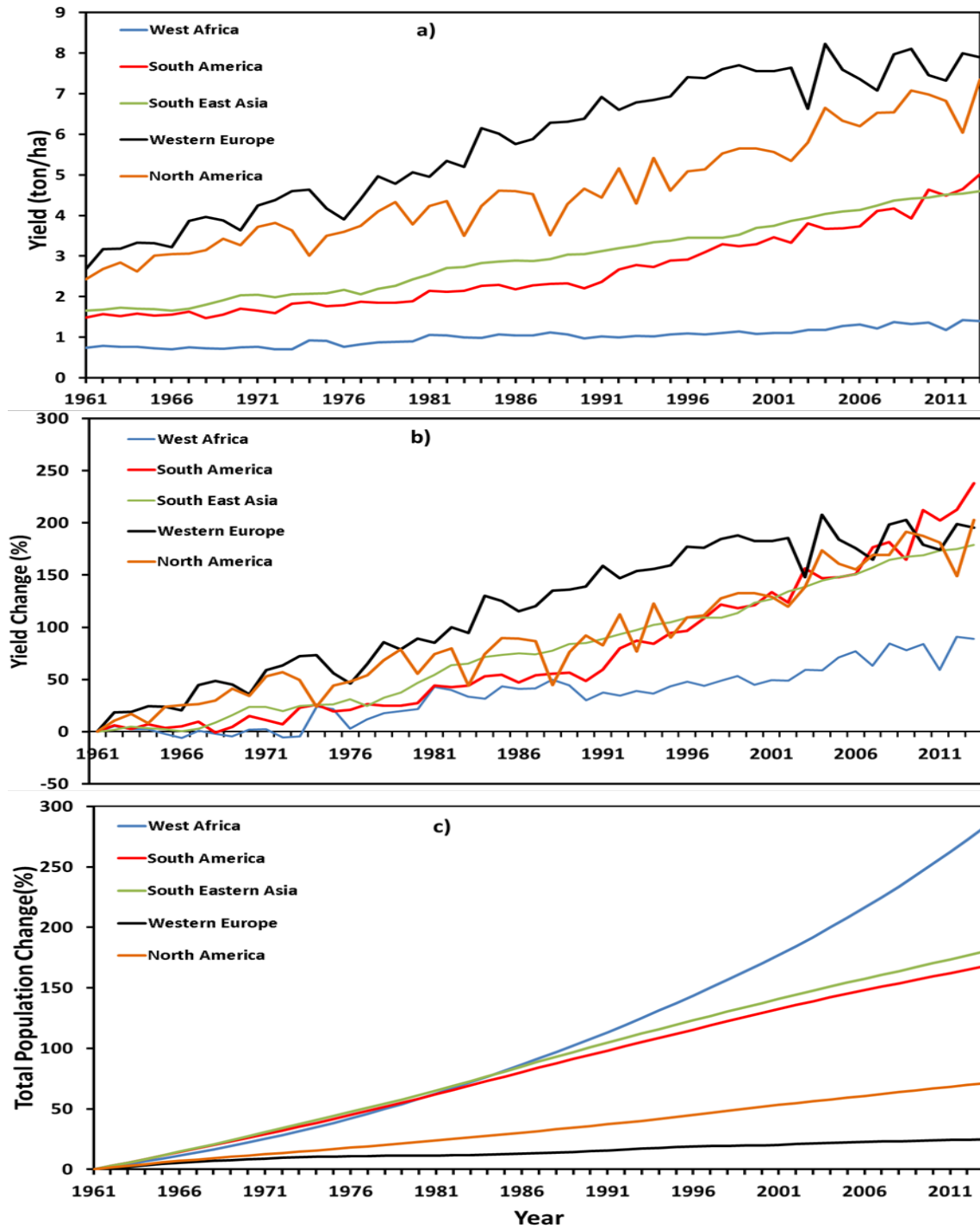


Figure S1: Cereals yield (a, b) and population(c) trend comparison. Source: FAOSTAT, 2017

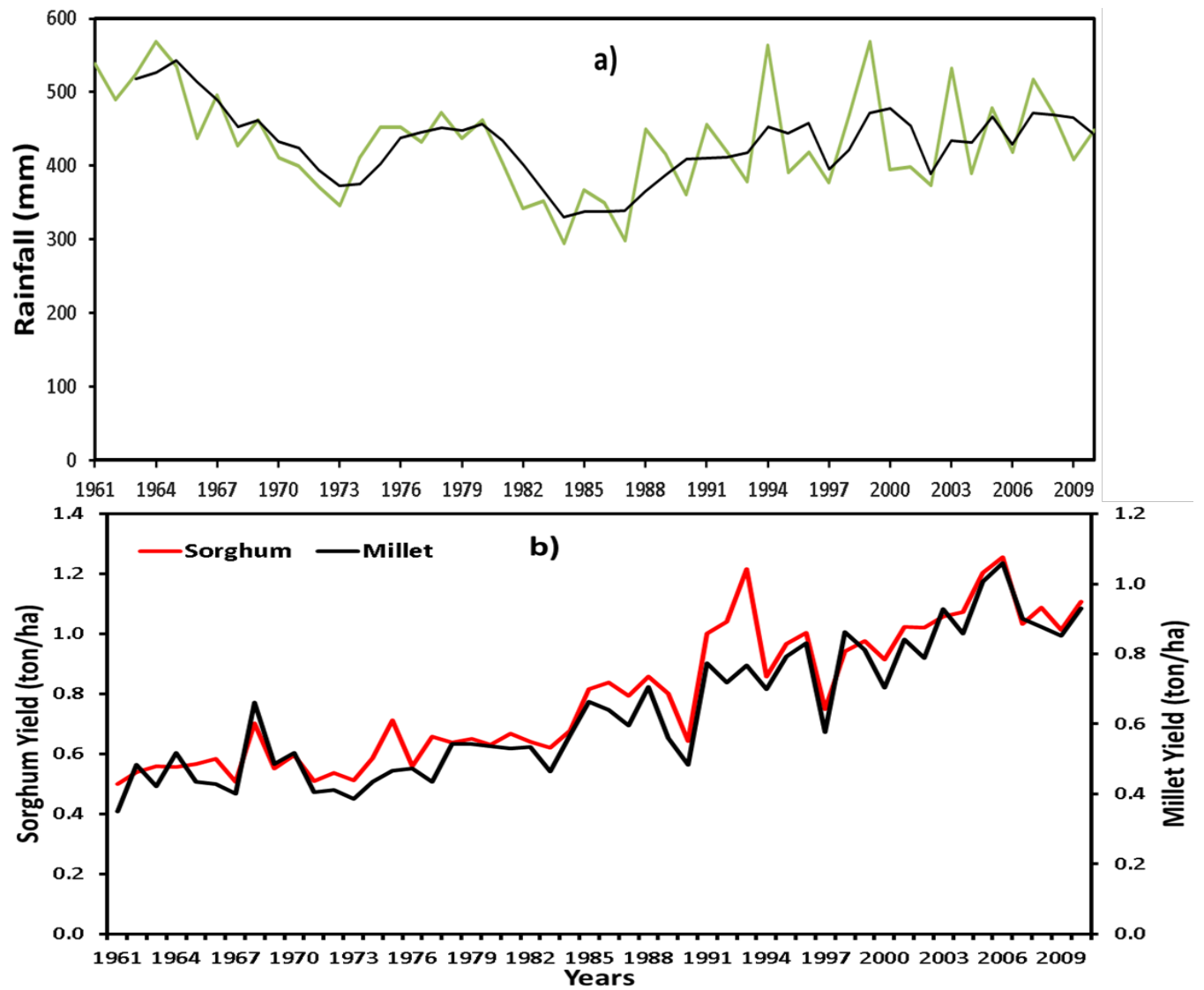


Figure S2. Decreasing rainfall (a) and increasing yields (b) at Dori in Burkina Faso

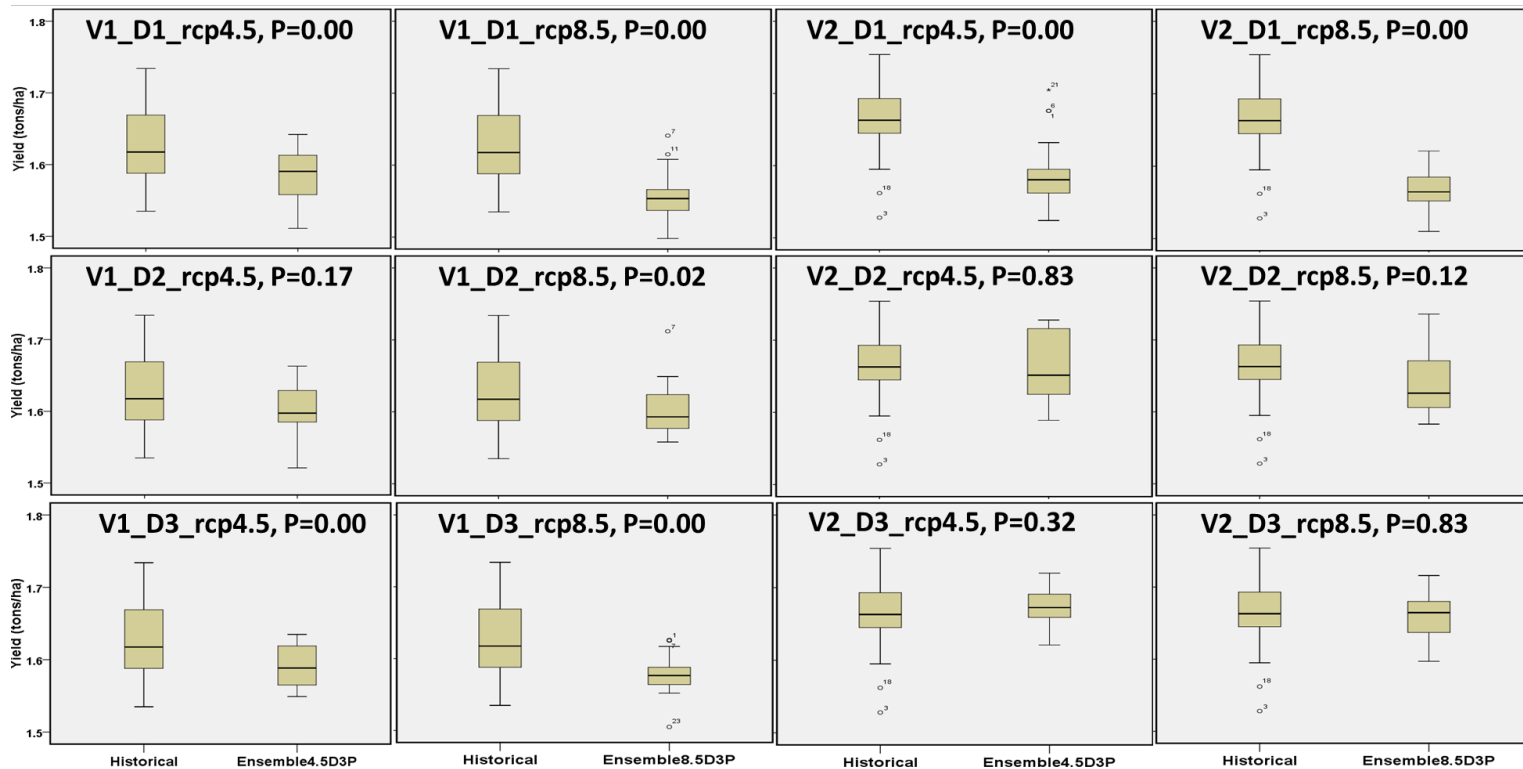


Figure S3: NTarla sorghum yield ensemble (9 GCM models) under the rcp4.5 and 8.5 for future period, 2021/25-2050 relative to the baseline periods, 1981/85-2010.

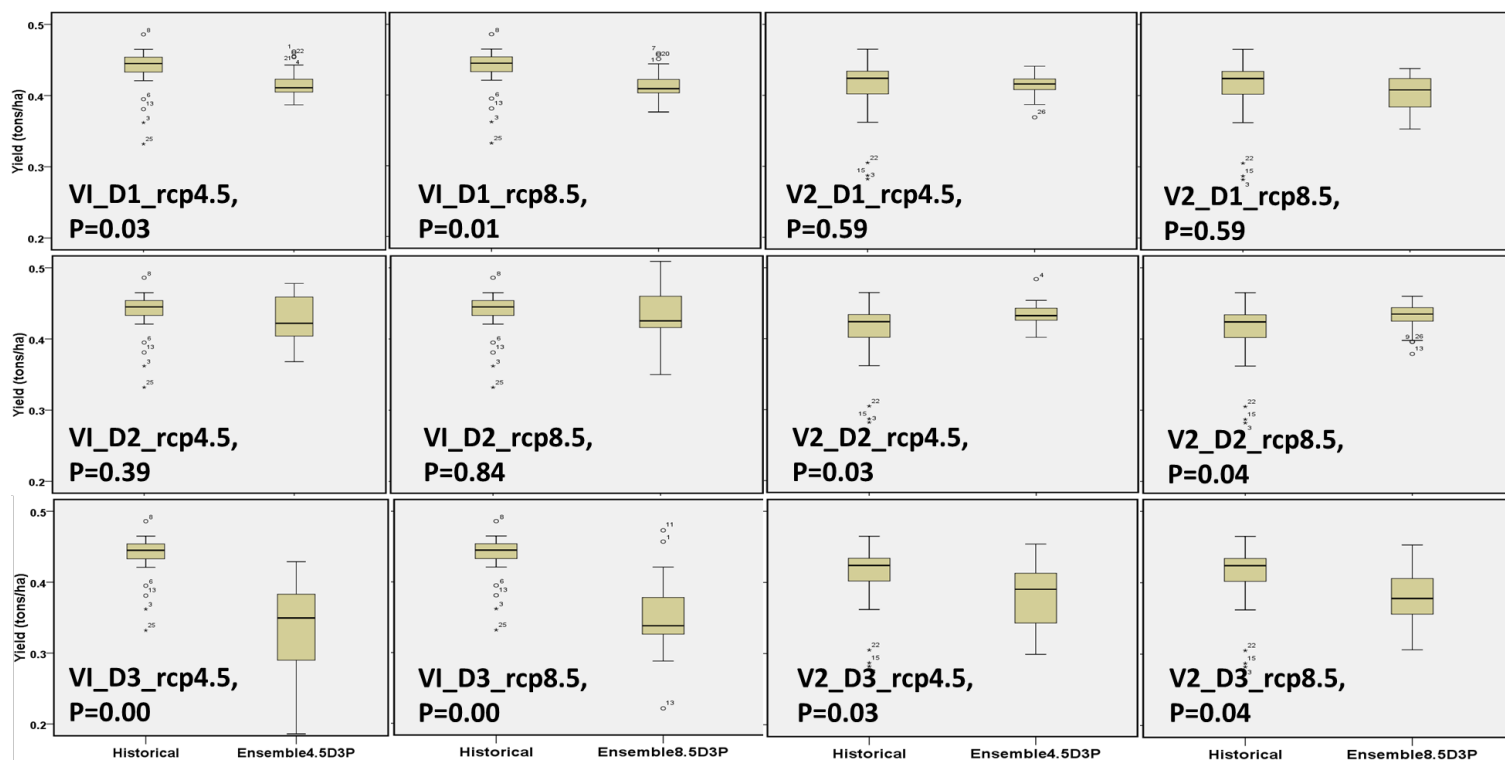


Figure S4: Tillabery millet yield ensemble (9 GCM models) under the rcp4.5 and 8.5 for future period, 2021/25-2050 relative to the baseline period, 1981/85-2010.

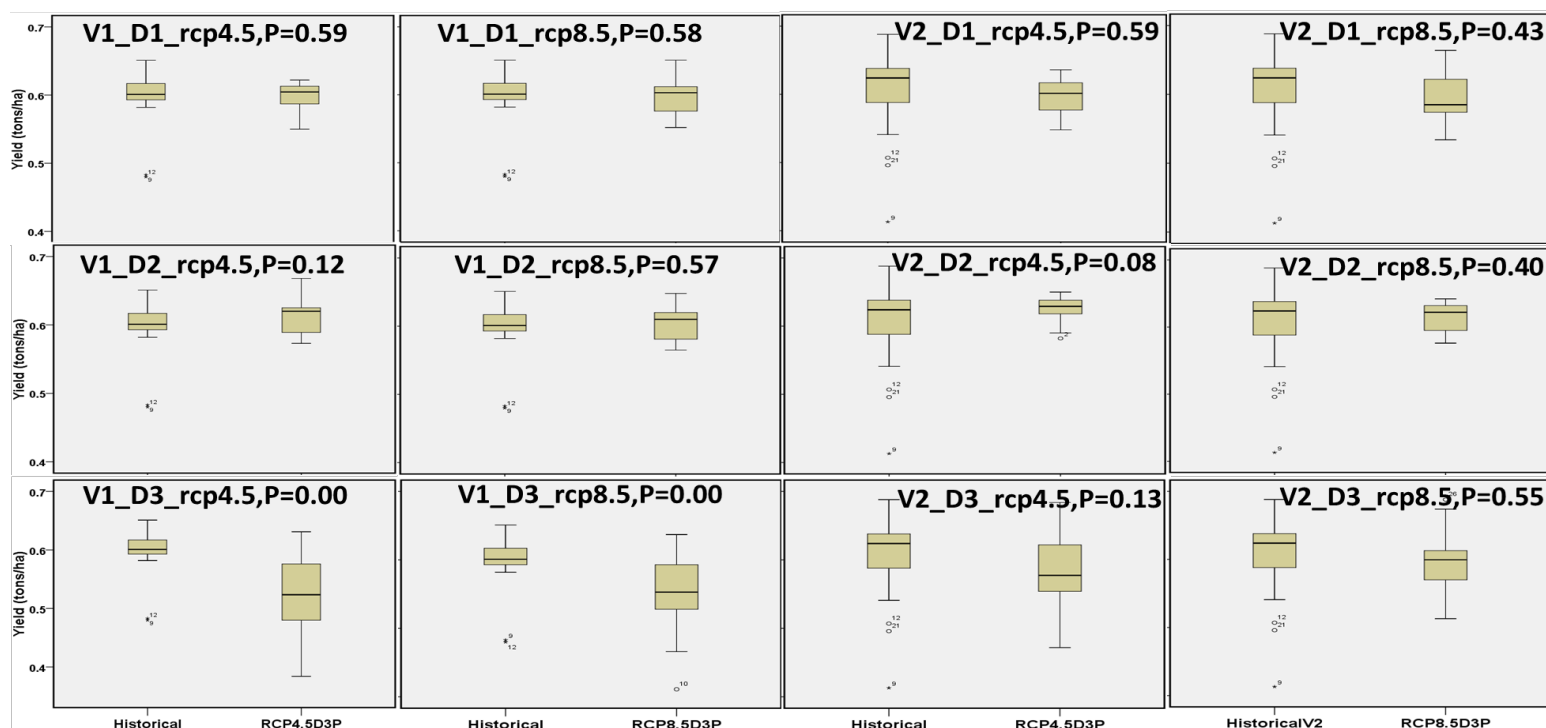


Figure S5: Dori millet yield ensemble (9 GCM models) under the rcp4.5 and 8.5 for future period, 2021/25-2050 relative to the baseline period, 1981/85-2010.

Supplementary Tables.

Table S1: Southern Guinea Savanna (Makurdi) average simulated maize and sorghum grain yield change for the future period (2025-2050) relative to the current period (1985-2010)

Scenarios	Yield(tons/ha) MaizeV1	YΔ(%)	Yield(tons/ha) MaizeV2	YΔ(%)	Yield(tons/ha) SorghumV1	YΔ(%)	Yield(tons/ha) SorghumV2	YΔ(%)
Histv1D1p	1.281	0	1.278	0	1.565	0	1.602	0
Rcp4.5V1D1P	1.307	2.1	1.329	4.0	1.572	0.4 ^a	1.605	0.2 ^a
RCP4.5V1D2P	1.313	2.5	1.340	4.9	1.630	4.1	1.668	4.2
RCP4.5V1D3P	1.329	3.7	1.344	5.1	1.647	5.2	1.693	5.7
RCP8.5V1D1P	1.317	3.3	1.335	4.4	1.562	-0.2 ^a	1.604	0.2 ^a
RCP8.5V1D2P	1.320	3.0	1.347	5.4	1.605	2.6	1.663	3.8
RCP8.5V1D3P	1.323	3.3	1.337	4.6	1.623	3.7	1.681	5.0
RCP8.5V1D1M	2.017	57.5	2.033	59.0	2.478	58.3	2.511	56.8
RCP8.5V1D2M	2.015	57.3	2.050	60.4	2.550	62.9	2.621	63.7
RCP8.5V1D3M	2.029	58.4	2.049	60.3	2.575	64.6	2.648	65.3
RCP8.5V1D1NP	2.791	117.9	2.832	121.6	3.522	125.0	3.481	117.4
RCP8.5V1D2NP	2.796	118.3	2.839	122.1	3.609	130.6	3.613	125.6
RCP8.5V1D3NP	2.804	118.9	2.834	121.7	3.690	135.8	3.685	130.1
RCP8.5V1D1OP	3.613	182.1	3.645	185.2	4.815	207.7	4.554	184.4
RCP8.5V1D2OP	3.590	180.3	3.660	186.4	4.892	212.6	4.735	195.7
RCP8.5V1D3OP	3.601	181.2	3.615	182.9	5.012	220.3	4.833	201.8

Note: All are significant at 0.05 except those specified as ^a

Table S2: Northern Guinea Savanna (Samaru illustrated) average simulated maize and sorghum grain yield change for the future period (2021-2050) relative to the current period (1981-2010)

Scenarios	Yield(tons/ha) MaizeV1	YΔ(%)	Yield(tons/ha) MaizeV2	YΔ(%)	Yield(tons/ha) SorghumV1	YΔ(%)	Yield(tons/ha) SorghumV2	YΔ(%)
Histv1D1p	1.628	0	1.628	0	1.360	0	1.397	0
Rcp4.5V1D1P	1.325	-18.6	1.393	-14.4	1.246	-8.4	1.257	-10.0
RCP4.5V1D2P	1.341	-17.6	1.348	-17.2	1.278	-6.0	1.276	-8.6
RCP4.5V1D3P	1.364	-16.2	1.385	-14.9	1.267	-6.9	1.303	-6.7
RCP8.5V1D1P	1.302	-20.0	1.303	-19.9	1.208	-11.2	1.239	-11.3
RCP8.5V1D2P	1.319	-19.0	1.321	-18.9	1.238	-8.9	1.253	-10.2
RCP8.5V1D3P	1.360	-16.5	1.368	-15.9	1.257	-7.6	1.287	-7.8
RCP8.5V1D1M	1.993	22.4	2.014	23.7	1.924	41.5	1.945	39.3
RCP8.5V1D2M	2.026	24.5	2.032	24.8	1.974	45.1	1.972	41.2
RCP8.5V1D3M	2.082	27.9	2.103	29.2	1.990	46.4	2.033	45.6
RCP8.5V1D1NP	2.758	69.4	2.801	72.0	2.724	100.3	2.682	92.0
RCP8.5V1D2NP	2.790	71.4	2.797	71.8	2.792	105.3	2.723	95.0
RCP8.5V1D3NP	2.908	78.6	2.921	79.4	2.835	108.4	2.812	101.3
RCP8.5V1D1OP	3.525	116.5	3.592	120.6	3.687	171.1	3.517	151.8

RCP8.5V1D2OP	3.595	120.8	3.597	121.0	3.577	163.0	3.577	156.1
RCP8.5V1D3OP	3.764	131.2	3.801	133.5	3.821	181.0	3.684	163.8

Note: All are significant at 0.05 level

Tablee S3: Sahelian zone (Dori and Tahoua illustrated) average simulated millet grain yield change for the future period (2021-2050) relative to the current period (1981-2010)

Dori/ Scenarios	Yield(tons/ha) V1	YΔ(%)	Yield(tons/ha) V2	YΔ(%)	Tahoua/ Scenarios	Yield(tons/ha) V1	YΔ(%)	Yield(tons/ha) V2
Histv1D1p	0.601	0	0.604	0	Histv1D1p	0.412	0	0.402
Rcp4.5V1D1P	0.597	-0.7ab	0.598	-1.0ab	Rcp4.5V1D1P	0.411	-0.1ab	0.410
RCP4.5V1D2P	0.614	2.2ab	0.624	3.4	RCP4.5V1D2P	0.417	1.2ab	0.432
RCP4.5V1D3P	0.524	-12.9	0.582	-3.6ab	RCP4.5V1D3P	0.333	-19.2	0.379
RCP8.5V1D1P	0.596	-0.8ab	0.594	-1.6ab	RCP8.5V1D1P	0.400	-2.8ab	0.409
RCP8.5V1D2P	0.606	0.8ab	0.614	1.6ab	RCP8.5V1D2P	0.415	0.7ab	0.420
RCP8.5V1D3P	0.553	-7.9	0.596	-1.3ab	RCP8.5V1D3P	0.353	-14.1	0.395
RCP8.5V1D1M	0.922	53.4	0.871	44.2	RCP8.5V1D1M	0.612	48.7	0.602
RCP8.5V1D2M	0.933	55.3	0.887	46.9	RCP8.5V1D2M	0.633	53.7	0.620
RCP8.5V1D3M	0.687	14.4	0.793	31.3	RCP8.5V1D3M	0.603	46.5	0.480
RCP8.5V1D1NP	1.259	109.5	1.219	101.9	RCP8.5V1D1NP	0.832	102.2	0.831
RCP8.5V1D2NP	1.275	112.1	1.246	106.3	RCP8.5V1D2NP	0.855	107.9	0.846
RCP8.5V1D3NP	0.721	20.0	0.968	60.2	RCP8.5V1D3NP	0.739	79.6	0.495
RCP8.5V1D1OP	1.601	166.4	1.524	152.4	RCP8.5V1D1OP	1.048	154.8	1.017
RCP8.5V1D2OP	1.607	167.3	1.610	166.6	RCP8.5V1D2OP	1.062	158.1	1.069
RCP8.5V1D3OP	0.746	24.1	1.109	83.6	RCP8.5V1D3OP	0.824	100.3	0.495

Note: ab indicate not Significant at 0.05